Assignment 3

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Fixed-Effect and Random-Effects Meta-Analysis

The data comes from the FDA meta-analysis of pediatric antidepressant randomized clinical trials. There are 24 studies comparing antidepressant treatment with placebo, with the outcome being the number of suicide-related events, including both thoughts and behaviors. The risk ratio (RR) here is defined as

$$\text{RR} = \frac{\mathbb{P}\left(\text{suicidal event}|\text{drug}\right)}{\mathbb{P}\left(\text{suicidal event}|\text{placebo}\right)},$$

with the "crude" / unadjusted risk ratio being ≈ 1.94, with a risk ratio greater than 1 suggesting a higher risk under antidepressant treatment. I fit a fixed-effect model using the Mantel-Haenszel estimator and a random-effects model using the DerSimonian-Laird moment estimator of between-study variance.

Model Estimates

Under the fixed-effect model, we assume that the studies share one true treatment effect, with the observed differences between studies arising from sampling variation. The Mantel-Haenszel method then combines the study-specific information to estimate this common risk ratio.

The random-effects model instead allows the true treatment effect to differ between studies. The DerSimonian-Laird method uses a moment-based estimate of the between-study variance, τ^2, and combines the study estimates while accounting for this extra source of variation.

Table 1: Pooled Treatment Effects from the Two Meta-Analysis Models

Model	Risk Ratio	95% CI	p-value
Fixed effect (Mantel-Haenszel)	1.84	(1.20, 2.81)	0.005
Random effects (DerSimonian-Laird)	1.66	(1.05, 2.61)	0.029

Both models suggest a higher risk of suicide-related events for patients receiving antidepressants compared with placebo. Under the fixed-effect model, the estimated risk ratio is 1.84 (95% CI: 1.20 to 2.81). Under the random-effects model, the estimated risk ratio is 1.66 (95% CI: 1.05 to 2.61). Both confidence intervals exclude 1, suggesting that the risk of suicide-related events was higher for patients assigned to antidepressants in these trials.

Heterogeneity and Model Choice

To assess whether the treatment effects differ across studies beyond what we would expect from sampling variation, we use Cochran's Q test. The null hypothesis of the Q test is that the studies have a common true treatment effect, ie $\tau^2 = 0$. The test statistic is

$$Q = \sum_{i=1}^{k} \omega_i (y_i - \bar{y}_\omega)^2,$$

where y_i is the estimated log risk ratio for study i, and ω_i is the weight assigned to study i. With 20 studies that contain nonzero suicidal events in both drug and placebo groups, we have 19 degrees of freedom for this Q test.

The heterogeneity test gives $Q(19) = 10.68$, with $p = 0.934$. Thus, the data do not provide statistically significant evidence of between-study heterogeneity.

However, one important limitation is that the outcome is very rare. Bhaumik et al. (2012) show that moment-based estimates of treatment effects and heterogeneity can be biased in sparse binary data, and standard heterogeneity tests may fail to detect variation that actually exists. In this case, a nonsignificant Q test is therefore not especially strong evidence that the true effects are identical across studies.

If the common-effect assumption were justified, the Mantel–Haenszel estimate would be a reasonable summary of these trials. A random-effects model is also scientifically plausible because the studies involve different antidepressants. However, the DerSimonian–Laird estimate of τ^2 has its own limitations with rare events. Therefore, neither model is definitively correct. Rather, the two models together as a useful comparison rather than choosing a model solely from the heterogeneity test.

Forest Plot

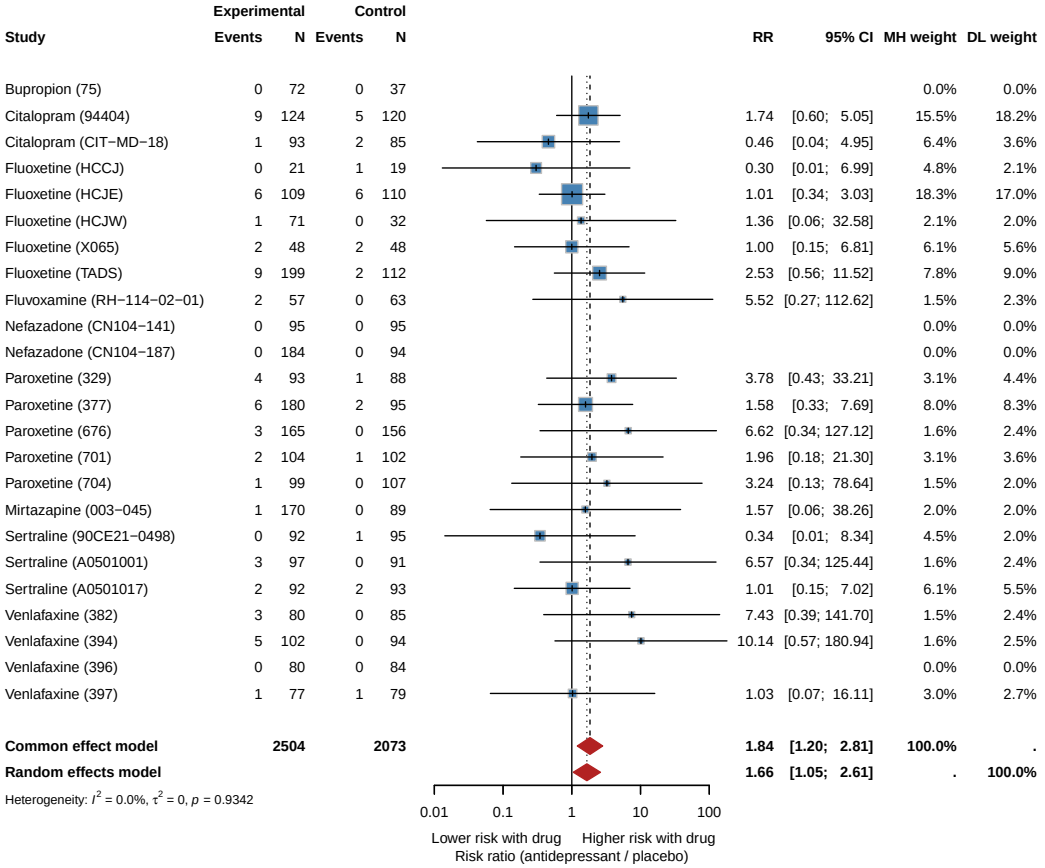


Figure 1: Forest plot of risk ratios for suicide-related events in antidepressant versus placebo groups.

The forest plot shows that many individual trials have wide confidence intervals, reflecting the small number of suicide-related events in each study. For studies with a zero-event cell in only one arm, a 0.5 continuity correction was used to obtain finite estimates.

Likelihood-Based Random-Effects Meta-Analysis

Since each observation is an event count out of a known number of patients, binomial mixed-effects logistic regression is an appropriate model for this analysis. There are 24 trials, each with a drug arm and a placebo arm. Let *study* be the grouping variable and let `Tx = 1` indicate treatment and `Tx = 0` indicate placebo.

Four random-effects structures were considered: a random study intercept, a random treatment slope, both effects assumed independent, and both effects allowed to be correlated. For the full model, let Y_{ij} be the number of suicide-related events in study i and treatment j, with n_{ij} patients in each treatment. The model is

$$Y_{ij} \mid b_{0i}, b_{1i} \sim \text{Binomial}(n_{ij}, p_{ij}), \qquad \text{logit}(p_{ij}) = \beta_0 + \beta_1 \text{Tx}_{ij} + b_{0i} + b_{1i}\text{Tx}_{ij}.$$

Here, β_0 is the average placebo log odds for a study with random effects equal to zero, and β_1 is the average treatment effect on the log-odds scale. The random intercept b_{0i} and random treatment slope b_{1i} represent study-specific departures from those values. The different candidate models are obtained by including or excluding these random terms and their covariance.

Model Comparison

Table 2: Comparison of Binomial Models with Different Random-Effects Structures

Model	Parameters	Log Likelihood	AIC
Random intercept	3	-78.28	162.56
Intercept + slope (independent)	4	-78.28	164.56
Intercept + slope (correlated)	5	-77.75	165.50
Random slope only	3	-85.29	176.57
No random effect	2	-89.39	182.79

The selected model is

$$\begin{aligned} Y_{ij} \mid b_{0i} &\sim \text{Binomial}(n_{ij}, p_{ij}), \\ \text{logit}(p_{ij}) &= \beta_0 + \beta_1 \text{Tx}_{ij} + b_{0i}, \\ b_{0i} &\sim N(0, \sigma_b^2). \end{aligned}$$

Table 3: Estimated Parameters for the Random-Intercept Binomial Model

Parameter	Estimate	SE	p-value
Intercept (β_0)	-4.682	0.279	< 0.001
Treatment (β_1)	0.710	0.239	0.003
Study intercept SD (σ_b)	0.783	–	–

The intercept, $\widehat{\beta}_0 = -4.682$, is the estimated log odds of a suicide-related event for a placebo patient in a study whose random intercept is zero. This is significantly below zero ($p < 0.001$), which is consistent with the low observed event rate. More importantly, the treatment coefficient is $\widehat{\beta}_1 = 0.710$ ($p = 0.003$). Holding the study-specific intercept constant, antidepressant treatment increases the estimated log odds of an event by this amount.

Exponentiating the treatment coefficient gives an odds ratio of 2.03, with an approximate 95% Wald confidence interval of (1.27, 3.25). Since the interval excludes 1, there is statistically significant evidence of a treatment effect in this model.

Estimated Probabilities of a Suicide-Related Event

For a study with random intercept $b_{0i} = 0$,

$$\text{Placebo log odds} = \hat{\beta}_0,$$
$$\text{Drug log odds} = \hat{\beta}_0 + \hat{\beta}_1.$$

If η denotes the log odds, the corresponding odds are e^{η}, and the event probability is

$$p = \frac{e^{\eta}}{1 + e^{\eta}}.$$

Table 4: Estimated Event Probabilities for a Study with Random Intercept Zero

Group	Log Odds	Odds	Probability
Placebo	-4.682	0.0093	0.92%
Antidepressant	-3.973	0.0188	1.85%

The estimated event probability is **0.92%** for placebo and **1.85%** for antidepressant treatment, when the study random intercept is zero.

AI Comparison

Prompt: *Here is the course textbook, related papers on meta-analysis, prompt, and dataset. Do problem 2, ie fit different forms of logistic mixed models, compare AIC and convergence, and interpret the fitted log odds, odds ratios, and event probabilities in the case where b_{0i} = b_{1i} = 0 (if applicable)*

In addition to comparing different random-effects structures, the AI also considered both logistic and probit mixed-effect models. The results were as follows:

Table 5: Comparison of Logistic and Probit Results

Quantity	logit	probit
Intercept	-4.682	-2.370
Treatment coefficient	0.710	0.300
Study intercept SD	0.783	0.325
Treatment p-value	0.003	0.002
Placebo probability	0.92%	0.89%
Drug probability	1.85%	1.92%
AIC	162.56	161.69

The probit coefficients are on a different scale from the logit coefficients, so their numerical sizes should not be compared directly. However, both link functions give very similar event probabilities, and both find the existence of a treatment effect. Furthermore, their AIC values are extremely similar to one another as well.

Marginal Versus Unit-Specific Probabilities

Recall the random-intercept binomial logistic model:

$$\begin{aligned} Y_{ij} \mid b_{0i} &\sim \text{Binomial}(n_{ij}, p_{ij}), \\ \text{logit}(p_{ij}) &= \beta_0 + \beta_1 \text{Tx}_{ij} + b_{0i}, \\ b_{0i} &\sim N(0, \sigma_b^2). \end{aligned}$$

Unit-Specific Probabilities

Recall the probabilities for a study with its random intercept set to zero, $b_{0i} = 0$. The estimated placebo and drug probabilities are

$$\hat{p}_{\text{SS},0} = \frac{e^{\hat{\beta}_0}}{1 + e^{\hat{\beta}_0}}, \qquad \hat{p}_{\text{SS},1} = \frac{e^{\hat{\beta}_0 + \hat{\beta}_1}}{1 + e^{\hat{\beta}_0 + \hat{\beta}_1}}.$$

Marginal Probabilities Using the Design Effect

We divide the fixed-effect linear predictor by

$$D = \sqrt{1 + k^2 \hat{\sigma}_b^2}, \qquad k = \frac{16\sqrt{3}}{15\pi}.$$

The estimated marginal probabilities can then be approximated by

$$\hat{p}_{\text{M},0} \approx \text{logit}^{-1}\left(\frac{\hat{\beta}_0}{D}\right), \qquad \hat{p}_{\text{M},1} \approx \text{logit}^{-1}\left(\frac{\hat{\beta}_0 + \hat{\beta}_1}{D}\right).$$

From the fitted model, $\hat{\beta}_0 = -4.682$, $\hat{\beta}_1 = 0.710$, and $\hat{\sigma}_b = 0.783$. Then, $D = 1.101$.

Table 6: Unit-Specific and Marginal Probabilities of Suicide-Related Events

Group	Unit-Specific	Marginal (Approx.)
Placebo	0.92%	1.40%
Antidepressant	1.85%	2.64%

Comparison and Interpretation

For placebo, the estimated probability increases from 0.92% in a study with random intercept zero to an approximate marginal probability of 1.40%. For antidepressant treatment, it increases from 1.85% to 2.64%.

Conceptually, the unit-specific estimates hold the study random effect fixed, whereas the marginal estimates account for the distribution of baseline risks across all studies. Because the logistic transformation is non-linear, averaging the probabilities is not the same as setting the random intercept to its mean of zero. This produces higher estimated marginal event probabilities for both groups. The treatment effect is also smaller on the marginal log-odds scale: the conditional odds ratio is $e^{\hat{\beta}_1} = 2.03$, compared with an approximate marginal odds ratio of $e^{\hat{\beta}_1/D} = 1.91$. This does not mean that averaging across studies removes the treatment effect.

References

Bhaumik, D. K., Amatya, A., Normand, S.-L. T., Greenhouse, J., Kaizar, E., Neelon, B., & Gibbons, R. D. (2012). Meta-analysis of rare binary adverse event data. *Journal of the American Statistical Association, 107*(498), 555–567.

Code

The complete R code used for this analysis can be found here:

GitHub Repository